

Improved Efficiency of Fingertip Reflective Pulse Oximetry with 2 positions Integrated Optical Biosensor

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ABSTRACT

Currently, the reflected-light fingertip blood oxygen meter has been used for medical purposes to diagnose the initial physical condition of patients due to the advantages of this equipment designed able to measure the amount of oxygen in the blood at other locations of the patient's body aside from finger positions such as the forehead, chest, wrists, etc. But a user still found the equipment limitations in measurement accuracy at these body positions compared with the measurement at the finger. This work proposes to develop a low cost newly designed reflected-light blood oxygen meter system to improve efficiency and accuracy of this equipment type when using at another position of the body and evaluate the system performance with comparison to commercial transmissive pulse oximetry. The newly designed system evaluated result by using student's t-test analysis observed prob > |t| greater than 0.05% at the 95% confidence intervals and gauge repeatability and reproducibility is 4.31% measurement variations to ensure that the newly designed system performance is comparable with a commercial pulse oximetry system and capable to accept in clinical condition.

KEYWORDS

pulse oximetry, SpO₂, Optical Biosensor

ACM Reference format:

Supawadee Worapruerkjaru, and Kitsada Dougjitjaroen. 2020. Improved Efficiency of Fingertip Reflective Pulse Oximetry with 2 positions Integrated Optical Biosensor. In *Proceedings of International Conference on Advances in Information Technology (IAIT2020)*, July 1-3, 2020, Bangkok, Thailand. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3406601.3406637>

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IAIT2020, July 1–3, 2020, Bangkok, Thailand
© 2020 Association for Computing Machinery.
ACM ISBN 978-1-4503-7759-1/20/07...\$15.00
<https://doi.org/10.1145/3406601.3406637>

1 Introduction

Pulse oximetry is a noninvasive method for accurately estimating blood oxygen saturation (SaO₂) by reading the peripheral blood oxygen saturation (SpO₂) using two different light sources from infrared and red light with photodiodes depending on measurement site [1], either the transmissive or reflective mode can be used. The transmissive mode, the light sources, and photodiode are each other opposite while reflective mode uses the light sources and photodiode on the same side. The transmissive is most used due to their accuracy and stability from the light sources have directly transmitted to the photodiode. This operation mode has implemented for the wrist-finger model that is very common in the market. The device is actually a finger model as it is placed on a finger which is then connected to a wrist display. The pulse oximetry on the finger is quite noticeable and the finger may not be too comfortable for a long period and continuous monitoring. Early work on the reflective mode of pulse oximetry has been carried out.

Liu Ying, et al.[2] designed a new SpO₂ reflectance pulse oximetry sensor to improve the quantity of the signal by the optimal source-detector space and a design new type of pulse oximetry sensor, which makes the detecting signal have the least dependence on changes in scattering properties of the tissue. The result observed many advantages of powerful photoelectricity signal and high Signal to Noise (SNR), the influence of unconscious movement reduces and considering the skin tissue as homogenous media is more reasonable and discusses the light path length distribution, and makes a revision to the formulation given by Lambert-Beer law. Hooseok, et al. [3] developed a reflectance pulse oximeter and addressed the practical issues and limitations associated with its use. They use designed oximeter to investigate the results of AC amplitudes and DC offsets of the red and infrared signals, including modulation ratio, which are critical factors to estimate SpO₂. Longmore, et al. [4] investigate a specific location on the human body where is the photoplethysmography (PPG) sensors simultaneously measure the heart rate, oxygen saturation in the blood(SpO₂), and respiration rate (RR) with a comparison to a commercial pulse oximeter. The results show that the forehead produced the most accurate results for HR and SpO₂ both at rest and walking, however, it had poor respiration results. Fontaine, et al. [5] design a reflective pulse oximeter prototype and used to measure SpO₂ and HR while sitting, standing, during self-induced hypoxia, and during self-induced hyperventilation to compare with the measurements taken by the commercial pulse

oximeter. The result observed accurately measure within an error of $\pm 1\%$ and $\pm 3\%$ for SpO₂ and PR respectively from the wrist while an error of $\pm 1\%$ and $+4\%$ for SpO₂ and PR respectively from the chest.

However, the demand for reflective pulse oximetry is continuously expanding due to it does not require a flat measurement site. It can be used at diverse measurement sites, for example, the chest, forehead, and wrists. In particular, if the wrist is the available measurement site, reflective pulse oximetry can be comfortably used in the form of a wrist band or wristwatch. But, the reflective pulse oximetry, particularly for monitoring oxygen saturation at the wrist, is currently not practiced clinically. In recent years, many reflective pulse oximeters have become commercially available products, but they are only for personal monitoring of oxygen saturation purposes.

In this study, the researcher shows how to develop a newly designed reflective pulse oximetry system with a comparison to a commercial transmissive pulse oximeter and summarizes all issues to discuss the feasibility of using the device for clinical purposes. The rest of this paper is ordered per detail as follows. Section 2 gives a SpO₂ measurement method in details, how does reflective pulse oximetry to measure SpO₂ and newly designed system description. Section 3 provides interpretation and evaluation of the result and section 4 makes a conclusion and introduces future work.

2 Methodology

2.1 SpO₂ measurement

Oximetry is based on the variation in the absorption of electromagnetic energy of the hemoglobin molecule when its chemical binding is altered. Oxyhemoglobin (HbO₂), deoxyhemoglobin (Hb), carboxyhemoglobin (HbCO), and methemoglobin (MetHb) have their own characteristic colors that can be described with extinction coefficients. The extinction coefficients of four most common hemoglobin species are presented in the wavelength range from 570 to 1000 nm the visible light end of the spectrum into the near-infrared and wavelengths most commonly used in pulse oximetry are 660 nm (red) and 940 nm (infrared). Their selection is based on the large difference in absorption of HbO₂ and RHb, manageable tissue absorption, and flatness of the spectrum. [6, 7] Likewise, SpO₂ is the amount of O₂ that is carried in the blood. In the human body, SpO₂ is defined as the ratio of HbO₂ to the total concentration of Hb (reduced Hb + HbO₂) present in the blood [8]

$$\text{Oxyhemoglobin Saturation (\%)} = \frac{(HbO_2)}{(Hb) + (HbO_2)} \times 100 \quad (1)$$

Hb is the iron-containing protein in red blood cells that transports O₂ to the tissues. This protein forms an unstable and reversible bond with O₂. Thus, Hb in its oxygenated state is called HbO₂ and exhibits a bright red color. Conversely, in its reduced state, it exhibits a purplish-blue color. [9]

The modern basis for determining the SpO₂ concentration using light sources and photosensor(s) was developed by Takuo Aoyagi and Michio Kishi in 1972. Using the Beer-Lambert law describes the attenuation of light with the properties of the material through which the light is traveling. According to the Beer-Lambert law,

$$I = I_0 e^{-\epsilon(\lambda)Cd}$$

Or

$$A = \ln\left(\frac{I_0}{I}\right) = \epsilon(\lambda)Cd \quad (2)$$

where A is the attenuation, I₀ is the incident light intensity, I am the received light intensity, $\epsilon(\lambda)$ is the molar extinction coefficient, C is the concentration of material, and d is the optical path length. The Beer-Lambert law enables us to measure SpO₂ by using the molar extinction coefficients of HbO₂ and RHb. Considering the molecule compound of tissue, Beer-Lambert law can be extended as follows:

$$A = d[\epsilon_{HbO_2}(\lambda)C[HbO_2] + \epsilon_{RHb}(\lambda)C[RHb] + \epsilon_{other}(\lambda)C_{other}] \quad (3)$$

Beer-Lambert law is valid when there is one unknown substance in a clear, non-turbid solution and constant path-length. In addition, there should be no chemical reactions or reactions between absorbent and solvent. [6, 7]

2.2 Reflective Pulse oximetry

Reflective pulse oximetry allows measurements to be taken from areas of the body in which transmissive pulse oximetry cannot be employed. The optical elements are thus placed on the same plane and the same body surface. Reflection originates from non-homogeneity in the optical path, i.e. at the interfaces between materials with different refractive indices. This means that on physiological grounds, strong reflections can be expected on the entry of light into the bone. The trans illuminated tissue must be well perfused to obtain as robust a signal as possible. It is not all human body parts, as well as perfused the anger or hands, but an AC/DC ratio of 0.001 to 0.005 can be obtained on the forehead. Perfusion is also good over the sternum. One example method of signal enhancement is to heat the measurement site to induce hyper perfusion, which can safely be performed up to 42 C. A rubefacient, e.g. nicotinic acid (Rubriment), can also be applied to the measurement site. The physical limitation of this oximetry type is direct light disturb, to increasing light intensity can make uncontrolled damage tissue at the measurement point and the wavelength of emitted light changes to be made LED warmer and incoming reflected light path is less well and may differ between the two wavelengths [10]. All limitation lists are important to make the system measurement accuracy problem. The drawbacks should be solved by establishing the cover sheet to shield the MAX30102 sensor from external light.

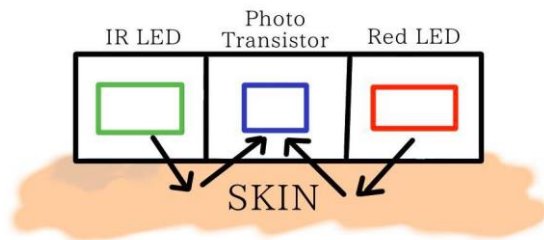


Figure 1: Reflective Pulse oximetry

2.3 System Design

The researcher has selected the MAX30102 chip of Maxim Integrated which integrated reflective pulse oximetry and heart-rate monitor module as an application circuit example in Figure 2. The application scenario of this chip has been used in the SAMSUNG Galaxy S7 smartphone and wearables. It includes internal LEDs, photodetectors, optical elements, and low-noise electronics with ambient light rejection. The MAX30102 provides a complete system solution to ease the design-in process for mobile and wearable devices. The SpO₂ subsystem of the MAX30102 contains ambient light cancellation (ALC), a continuous-time sigma-delta ADC, and a proprietary discrete-time filter. The ALC has an internal Track/Hold circuit to cancel ambient light and increase the effective dynamic range. The SpO₂ ADC has programmable full-scale ranges from 2μA to 16μA. The ALC can cancel up to 200μA of ambient current. The internal ADC is a continuous-time oversampling sigma-delta converter with an 18-bit resolution. The ADC sampling rate is 10.24MHz. The ADC output data rate can be programmed from 50 SPS (samples per second) to 3200sps. The designed system represents in Figure 3 below.

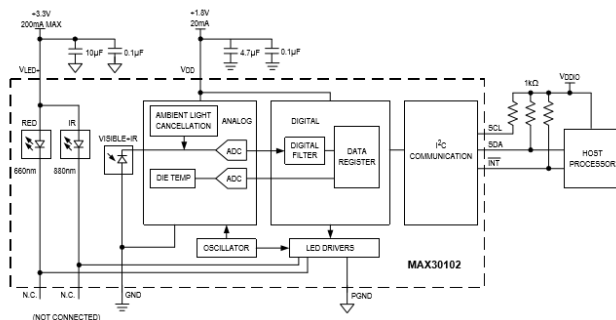


Figure 2: MAX30102 Application Circuit Example

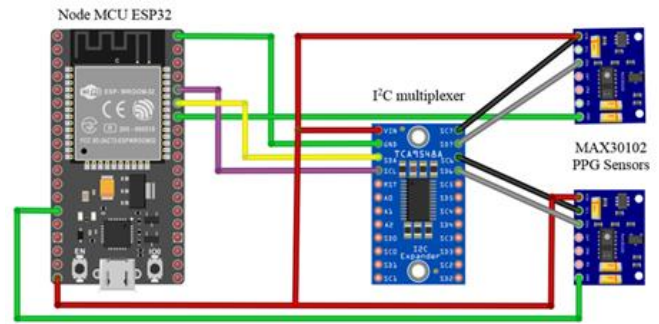


Figure 3: New designed SpO₂ sensor

The newly designed system consists of 2 units of MAX30102 to connect with the Node MCU ESP32 micro-controller via I²C bus pass through I²C multiplexer. The ESP32 is ultra-low power single chip 2.4 GHz WiFi and Bluetooth combo designed with the TSMC 40 nm. technology for mobile, wearable electronics, and Internet of Things (IoT) applications from ESPRESSIF Inc. It features all the characteristics of low-power chips, including fine-grained clock gating, multiple power modes, and dynamic power scaling. With dual-core 32-bit/160 MHz/512K SRAM/36 GPIO/12 bits ADC and 2 of the I²C bus are more powerful to support this new system design. The TCA9548A low voltage 8-channel I²C multiplexer switch from Texas Instrument is used to wire up the two I²C MAX30102 SpO₂ sensor to ESP32 due to this chip has a fixed I²C address, and from what we cannot have two devices with the same address on the same SDA/SCL I²C pins. It has eight bidirectional translating switches that can be controlled through the I²C bus, 3 address pins that allowing up to eight TCA9548A devices on the I²C bus and ESD protection exceeds JESD22. The 2 sensors which are put into the system are proposing to improve the accuracy of SpO₂ measurement by using the average of raw data from 2 sensors. The SpO₂ raw data from these 2 sensors should be processed by mode MCU and report measurement SpO₂ for the user to know their oxygen saturation in the blood. The SpO₂ and HR calculation process are successfully done to follow the pseudocode as follows.

Setup Process:

```
Select I2C MUX Channel #1
Initial and setup MAX30102 sensor#1
Select I2C MUX Channel#2
Initial and setup MAX30102 sensor#2
```

Calculation Process:

```

While (1)
  Set/clear BUFFER#1
  Set/clear BUFFER#2
  Check MAX30102 sensor#1,2 status
  Get RED, Get IR of sensor#1
  Get RED, Get IR of sensor#2
  Select I2C MUX Channel #1
  Calculate SpO2/HR of sernsor#1
  Select I2C MUX Channel #2
  Calculate SpO2/HR of sernsor#2
  Next sample
End

```

3 Result and Discussion

Before starting to collect SpO₂ raw data, the setting up MAX30102 process to work well at optimal mode (High response time and accuracy) is required. The setting parameters are light source brightness, sample average, sample rate, pulse width, and ADC average. The C/C++ coding has been developed and used for finding and setting up these parameters based on manufacturing recommended configuration and operating profile for MAX30102 EV kits.[11] The optimal value that is used to set up these parameters are Table 1

TABLE 1. The MAX30102 Setting Value

Setting parameter	Value
Light source brightness	10 mA
Sample average	8
Sample rate	100 Hz
Pulse width	411 Hz
ADC average	4096 (12 bits)

After that, the researcher has collected SpO₂ raw data from 3 patients with a newly designed system totally 20 data points (3 times per data points) per person and average raw data of the 2 optical biosensors to be the unique data point and compare it with selected commercial fingertip oximetry. Figure 4 shows the box plot graphical summary comparison of the newly designed system which has a standard deviation (std dev) higher than commercial oximetry while the normal quantile plot of a newly designed system that approximates a blue diagonal straight line is normally distributed. So, the student's t-test analysis to compare SpO₂ performance between newly designed (μ_{new}) and commercial transmissive pulse oximetry (μ_{com}) comparison based on the null hypothesis $P(H_0: \mu_{new} = \mu_{com})$, observed p-value = 0.524 (prob > |t| in Figure 4 based on JMP software descriptive statistical) greater than 0.05% (95% confidence)

which should be accepted null hypotheses. So, the newly design system performance is comparable with the commercial transmissive pulse oximetry system per expectation.

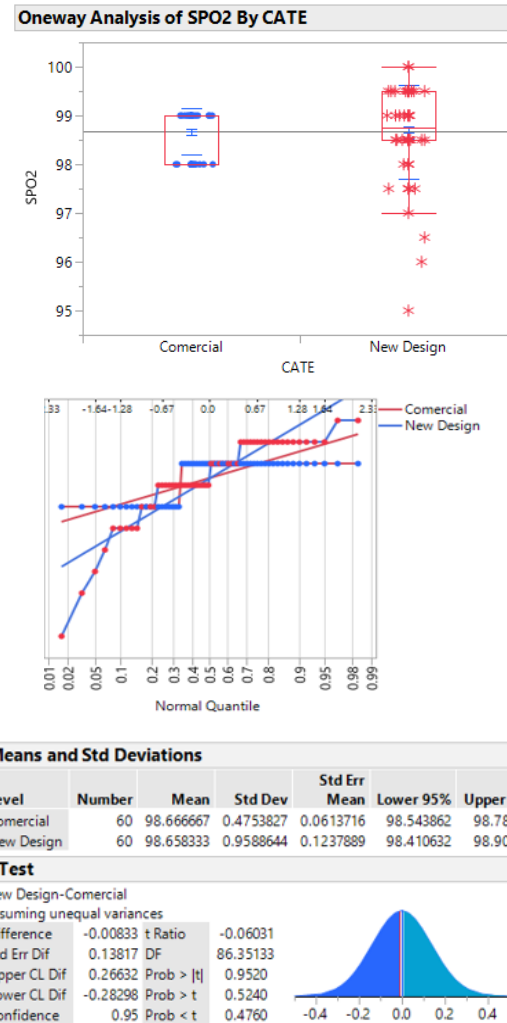


Figure 4: One Way Analysis of newly designed system performance

Moreover, the researcher evaluates the measurement system capable of a newly designed reflective oximetry system to measure gage repeatability and reproducibility as the result in Figure 5. Reproducibility is a little impact on the total system variation that the majority comes from the 3 patients differentiate caused by using a single device used. When substituting these variations into the equation, the total GR&R result of the newly designed system is 4.31% measurement variations. So, the measurement capability of the new design is excellent to accept clinical conditions.

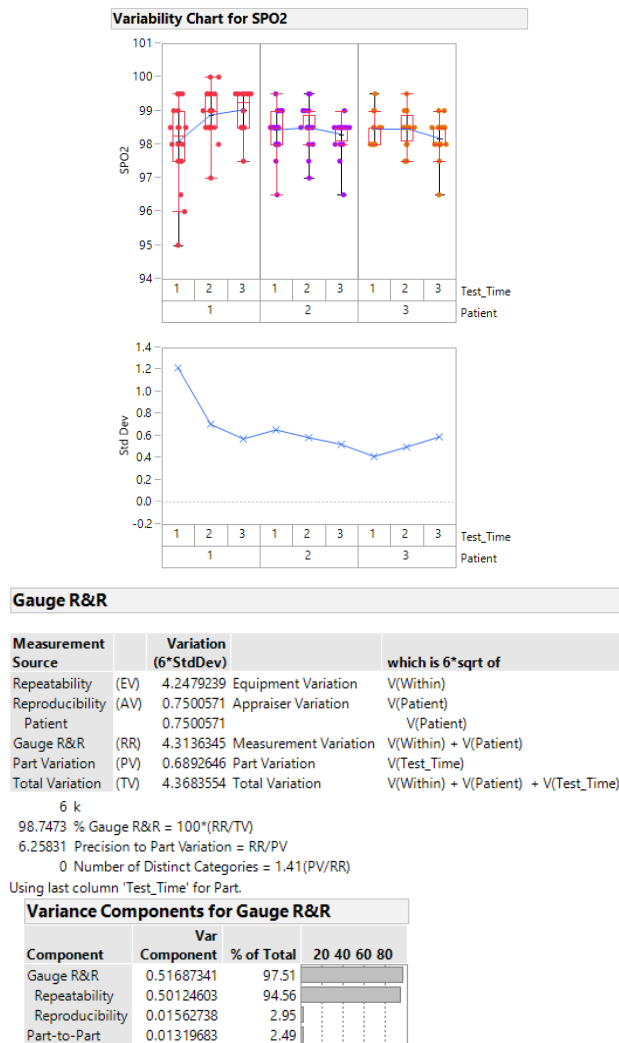


Figure 5: The GR&R Analysis of newly designed system performance

4 Conclusion and Future works

Per the experiment result, the standard deviation of the newly designed system is also pretty large than the commercial system. It is due to some factor that should highly contribute with SpO2 reading for example the cover sheet has no 100% efficiency to shield the sensor from an external light source and the pressing force from the patient's finger to place on SpO2 sensor is not smoothly enforce. So, the cover sheet and the pressing force are the main factors to considering for the reflective pulse oximeter design to implement it in the clinical conditions.

The future work is proposing to leverage this newly designed reflective pulse oximeter to measure blood oxygen saturation at other body locations for example wrists, chest, forehead, etc., and feed the SpO2 and HR raw data to the cloud for remotely

monitoring. Due to the advantage of the newly designed system that integrated IoT based node MCU and 2 positions integrated optical biosensor reflective pulse oximetry has designed to increase the efficiency of this equipment beyond the limitation.

ACKNOWLEDGEMENTS

The work acknowledges the help provided by the research advisor and technical staff at Darunsikkhalai Science School, King Mongkut's University of Technology Thonburi for their kind support.

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